

A Introduction

Aerial 3D scene understanding is essential for autonomous UAV operations such as cargo delivery, search-and-rescue, and human-UAV collaboration. In these settings, an autonomous platform must not only recognize visible image regions, but also infer metric geometry, free space, and the semantic occupancy of partially observed volumes. Such a representation is important for collision avoidance, viewpoint planning, landing-zone assessment, and interaction with objects and humans in cluttered outdoor environments. Camera-based perception is particularly attractive for UAVs because RGB cameras are lightweight, inexpensive, widely available, and compatible with strict size, weight, and power constraints. However, inferring dense 3D structure from monocular or short video observations remains substantially harder than recognition, since the image formation process collapses scale, depth, and occlusion into a two-dimensional signal. Aerial video offers an additional source of geometry through motion parallax, but exploiting it requires more than generic temporal feature aggregation: the system must relate target and source views through camera calibration and relative pose.

Most progress in semantic scene completion (SSC) and camera-based 3D perception has been driven by indoor reconstruction and autonomous-driving benchmarks. Early SSC formulations used depth as input to jointly predict volumetric occupancy and semantics [15]. Later monocular methods, such as MonoScene [3], moved the problem toward RGB-only 2D-to-3D lifting, while VoxFormer [8], TPVFormer [6], SurroundOcc [17], OpenOccupancy [16], CGFormer [19], and FoundationSSC [4] developed increasingly capable view-transformation, voxel-attention, and semantic-geometric fusion designs. These works demonstrate that explicit 3D representations are valuable for scene-level reasoning, but their primary evaluation domains remain road scenes with ground-level perspective, relatively constrained camera height, and characteristic automotive object layouts. Aerial viewpoints violate many of these assumptions: altitude changes the apparent scale of all objects, oblique views produce mixed top-down and side-view evidence, fine semantic boundaries can be weakly textured, and occlusions differ from those observed by ground vehicles. Consequently, strong methods trained or designed for terrestrial scenarios do not necessarily transfer to UAV perception without careful adaptation.

OccuFly [5] creates an opportunity to study this gap in a real-world aerial setting. It provides RGB images, metric depth maps, camera calibration, and semantic occupancy grids for aerial scenes captured at multiple altitudes and across urban, industrial, and rural environments. Unlike earlier UAV semantic segmentation datasets such as UAVid [10], which focus on 2D dense labeling, or aerial reconstruction resources such as UrbanScene3D [9], OccuFly directly supports the joint study of metric depth estimation and 3D semantic scene completion from aerial camera observations. This makes it possible to ask whether modern visual foundation representations provide transferable geometric and semantic priors for aerial 3D scene understanding, rather than only for image classification, 2D segmentation, or ground-level depth prediction. It also enables

a more precise question: when multiple calibrated UAV views are available, does explicit multiview correspondence provide a stronger geometric prior for SSC than a monocular depth estimator alone?

In this work we investigate V-JEPA 2.1 [11] as a frozen foundation representation for aerial depth estimation and depth-aware semantic scene completion. V-JEPA and its successors learn by predicting representations in latent space rather than reconstructing pixels [1, 2]. V-JEPA 2.1 further emphasizes dense predictive supervision, deep self-supervision at intermediate layers, and unified image/video tokenization, making it a natural candidate for dense 3D perception. The central question of this paper is not whether a large foundation model performs well on generic vision tasks, but whether its spatial and spatio-temporal features encode the aerial geometry needed for metric depth and volumetric semantic completion. We therefore keep the backbone frozen and vary the decoder, metadata conditioning, temporal aggregation, and geometric lifting mechanism.

We study this question along three axes. First, for monocular depth estimation, we compare frozen linear probes against structured DPT-style decoders [14]. The linear probes test how linearly accessible metric depth is from V-JEPA 2.1 features, while the DPT decoders test whether multiscale token reassembly and convolutional fusion can recover fine spatial detail from frozen transformer features. We also examine a Simple Feature Pyramid (SFP) adapter inspired by ViTDet [7], which converts a single final-layer feature map into a pyramid suitable for dense prediction. This allows us to study the dense final-layer representation produced by V-JEPA 2.1, which, according to its authors, incorporates information from earlier layers into the final feature map. Because aerial scale is strongly coupled to flight altitude and camera calibration, we further condition selected decoders with feature-wise linear modulation (FiLM) [12] using altitude and camera intrinsics. Second, we compare spatial RGB features with spatio-temporal video features. Aerial videos contain motion parallax and temporal continuity that may help resolve depth ambiguities.

Third, for SSC, we compare monocular depth priors inside depth-aware lifting architectures inspired by Lift-Splat-Shoot [13], CGFormer [19], and FoundationSSC [4]. The monocular V-JEPA/FoundationSSC branch uses a DPT-FiLM depth-probability frontend to lift semantic context into the OccuFly voxel grid. We further introduce a Pose-MVS-Cost-Volume FoundationSSC variant that replaces the monocular depth-probability frontend with a posed multiview stereo cost volume in the spirit of learned MVS cost-volume methods [18]. Given a target UAV view and neighboring calibrated source views, this branch matches target/source V-JEPA video features across metric depth hypotheses, regularizes the resulting cost volume, and converts it into the depth distribution consumed by the FoundationSSC lifting module. We also evaluate a Pose-MVS-FiLM variant that applies altitude/intrinsics FiLM conditioning directly to the MVS depth-bin logits before the depth softmax, so that the dense depth map and lifting probabilities remain consistent.

This design separates two different uses of video. Dense temporal attention tests whether frozen V-JEPA video tokens contain useful target-frame depth,

semantic contexta and world reasoning cues. In contrast, Pose-MVS tests an explicitly geometric hypothesis: calibrated source views should constrain depth through projective correspondence, even when monocular appearance cues are ambiguous. This distinction is important for aerial perception, where altitude variation, repeated textures, vegetation, roofs, and oblique viewing directions can make single-image depth uncertain while neighboring UAV frames may provide exploitable parallax.

Our contributions are as follows:

- We present a systematic study of frozen V-JEPA 2.1 spatial and spatio-temporal representations for aerial metric depth estimation on OccuFly.
- We compare representation probes of increasing structure: linear depth heads, DPT-style multiscale decoders, SFP-adapted final-layer decoders, and altitude/intensity-conditioned FiLM variants.
- We analyze the role of spatio-temporal features through dense temporal-attention adapter.
- We extend the investigation from depth to semantic scene completion by incorporating predicted depth and depth distributions as geometric priors in CGFormer-, and FoundationSSC-inspired aerial SSC variants.
- We introduce Pose-MVS-Cost-Volume FoundationSSC, which replaces monocular depth-probability lifting with pose-aware multiview cost-volume depth estimation from target/source UAV views.
- We further evaluate Pose-MVS-FiLM, a metadata-conditioned MVS variant that modulates cost-volume depth logits with altitude and camera intrinsics while preserving probability depth consistency for voxel lifting.
- We identify the practical domain gap between ground-level camera-centric 3D perception and aerial 3D scene understanding, providing a focused evaluation protocol for future foundation-model transfer studies in UAV perception.

References

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